

Navigating the IQ Triangle in Transformers

Topic

Architectural Trade-offs and Expressivity

Background

Transformer architectures are inherently bounded by the constraints of the IQ Triangle, which dictates a strict trade-off between Expressivity, Theoretical Complexity, and Training Parallelism.

Assignment Task

Analyze the constraints of the IQ Triangle within the context of standard Transformer models. Given that maximizing theoretical efficiency and training parallelism often bottlenecks representational power, propose and defend structural modifications or algorithmic strategies designed to specifically elevate the model's expressivity without breaking the practical limits of the other two vertices.

Submission Expectation

Prepare a rigorous, self-contained written response that defines all key assumptions, uses precise technical terminology, and supports the argument with mathematical, architectural, or conceptual reasoning where appropriate.